



TOFU for Task Offloading in UAVaaS

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Reference

Kumar, Prince, Veera Manikantha Rayudu Tummala, and Arijit Roy. 2025. "TOFU: Trust-Based Task Offloading in UAV-as-a-Service." IEEE Transactions on Vehicular Technology, September 01, 2025.

Background

- Integration of UAVs with IoT architecture
- Services:
 - Surveillance
 - Environmental modelling
 - Delivery of goods
- Proposal of UAVaaS
- One UAV → multiple heterogeneous sensors
- A UAV can process multiple tasks of same or different applications
- Single UAV may not be able to serve all tasks of an application
- Multiple UAVs collaborate to execute all tasks and complete application

The Problem of Task Offloading

- Scenario: A UAV may not have adequate resources to complete its assigned task
- What can be done: Offloading that task to another UAV
- Problem: How to offload the task to a trustworthy UAV that is capable of completing the given task
- Proposed solution: TOFU (Trust-Based Task Offloading in UAV-as-a-Service)

Introduction to TOFU

- During task offloading, a resource-constrained UAV needs to offload the task to a UAV with adequate resources
- However, not all available UAVs with adequate resources are trustworthy enough to offload the task
- It is essential to design a mechanism to evaluate the trustworthiness of a UAV
- Proposed scheme TOFU determines the trustworthiness of a UAV by using a debit-credit score approach

UAV with
limited resources

offload a task

UAV with
adequate resources

but is it trustworthy?

evaluate trust

how? using TOFU.

residual
energy

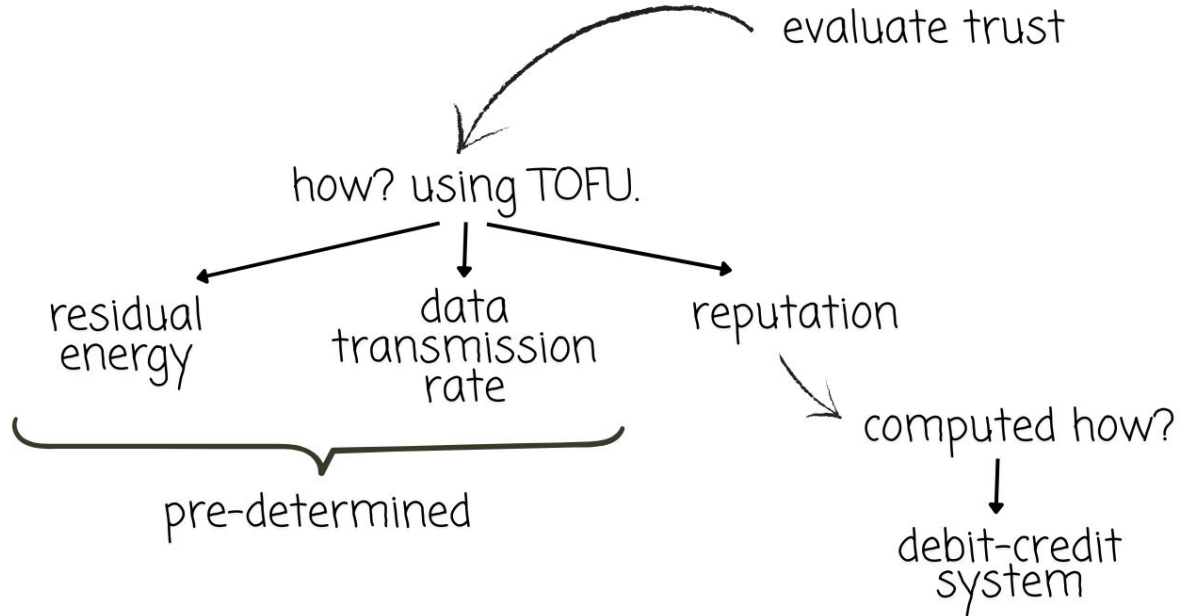
data
transmission
rate

reputation

pre-determined

computed how?

debit-credit
system



Introduction to TOFU

- For a UAV, TOFU evaluates trust as a function of 3 components:
 - Residual energy
 - Data transmission rate
 - Reputation
- We compute the reputation of a UAV as follows
 - The resources available with a UAV are considered to determine the initial reputation
 - A debit-credit system is used to compute the updated reputation value
- Reputation is transferred between UAVs when a task is offloaded
- Rubinstein Bargaining Model
- Subgame Perfect Nash Equilibrium (SPNE) to decide optimal amount of reputation being transferred between UAVs

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- Definitions -
 - Constraints -
 - Problem Statement -
 - Methodology -

UAVs and Tasks

- Set of UAVs, $U = \{U_1, U_2, U_3, \dots, U_n\}$
- Set of tasks, $T = \{T_1, T_2, T_3, \dots, T_k\}$
- Where $T_i = \langle t_i, U_i \rangle$
 - $t_i \rightarrow$ the task
 - $U_i \rightarrow$ the UAV it is assigned to
- In UAVaaS architecture, a few UAVs are idle, to which no task is allocated or offloaded
- Task UAVs (U^T) $\rightarrow$ which offloads the task
- Service UAVs (U^S) $\rightarrow$ which executes the offloaded task
- U^T and U^S are two disjoint sets
- $U^T \cup U^S \subseteq U$

Key Notations

Symbol	Description
U	Set of All the UAVs
U_i^T	i^{th} UAV assigned with a task
U_j^S	j^{th} UAV to which task is offloaded
ζ_{ij}	Trust between U_i^T and U_j^S
$\mathcal{E}_j$	Residual energy of U_j^S
$\mathcal{R}_i$	Reputation score of U_i
$\mathcal{D}_{ij}$	Data transfer rate between U_i^T and U_j^S
η_t	Parameter to decide if a task t is successfully executed
T_j^t	Time taken in executing task t by U_j^S
ET^t	Maximum allowed delay for task t
$\mathcal{A}_{ij}^t$	Award offered by U_i^T to U_j^S for completion of task t

Constraint 1

- A task UAV U_i^T offloads a task t to a service UAV U_j^S if and only if it is eligible in executing the task, and the eligibility of U_j^S is defined as:

$$e(U_j^S, t, U_i^T) = \begin{cases} 1, & \text{if } \zeta_{ij} \geq \zeta_t^{\min} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

- $\zeta_t^{\min} \rightarrow$ least required trust to offload the task
- $\zeta_t^{\min} = r \cdot (\tau \times (f_t \cdot M_t)) + d \cdot (\mathcal{D}_t^{\min}) \quad (4)$
 - $M_t \rightarrow$ Memory
 - $\mathcal{D}_t^{\min} \rightarrow$ Data rate requirements for a task, t
 - $f_t \rightarrow$ Minimum computational frequency required

Constraint 1

- $f_t = \frac{N_t}{T_t^{proc}}$ (3)
- $N_t \rightarrow$ number of instructions
- $T_t^{proc} \rightarrow$ maximum tolerable processing delay in a task, t

Constraint 2

- *At any given instant, there exists at least one UAV that is eligible to perform task t*
- *If no UAV possesses the resources that are required to execute a given task, the task is considered infeasible and is bound to fail. In such scenarios, the proposed method does not force an ineligible UAV to attempt the task, as doing so will affect its reputation*
- $\exists U_j^S \in U^S \text{ s.t. } e(U_j^S, t, U_i^T) = 1 \quad (5)$

Constraint 3

- *The mapping from task UAVs to service UAVs is always one-to-one, which indicates that every task UAV offloads its task to one and only one service UAV*
- $\forall U_i^T \in U^T, \exists \text{ one and only } U_j^S \in U^S$ (6)

Problem Statement

- Efficient task offloading:
 - transferring computational tasks from a task UAV to a service UAV
 - while executing them within the stipulated time
 - and ensuring the Quality-of-Service
- A task UAV must obtain a suitable service UAV that is capable of successfully executing the task
- Furthermore, a trust evaluation scheme that quantifies each UAV's trust in every other UAV is needed

Proposed Methodology

1. **Trust Evaluation**
2. Service UAV Selection
3. Dynamic Score Transfer (using Rubinstein Bargaining Game)

Trust Evaluation

- *Trust in the context of task UAV and service UAV is defined as the degree of confidence a task UAV has in the ability of a service UAV to complete the offloaded task successfully within the specified constraints*
- Trust between a task UAV, U_i^T and a service UAV, U_j^S is given as a function of:
 - Residual energy $\rightarrow \epsilon_j$
 - Data transmission rate of the channel between the 2 UAVs $\rightarrow D_{ij}$
 - Reputation, R_j

Trust Evaluation

- $\zeta_{ij} = \mathcal{E}_j \times (r \cdot \mathcal{R}_j + d \cdot \sigma(\mathcal{D}_{ij}))$ (1)
- $\mathcal{D}_{ij}$ is scaled to $[0,1)$ using sigmoid function
- $(r, d) \in [0, 1]$
- The initial trust value is bootstrapped using the 3 components
- Since trust cannot be directly observed, we assume a neutral baseline trust value when a task UAV offloads a task to a service UAV with no prior interaction
- This baseline is then adjusted dynamically based on the service UAV's performance in task execution, successful data transfers and energy availability

Residual Energy

- ϵ_j
- A UAV with higher residual energy is more likely to sustain the energy-intensive processes involved in task execution and data transmission

Data Transmission Rate

- Higher data rates facilitate faster offloading and task execution
- Shannon's channel capacity to model the maximum data transmission rate:

$$\mathcal{D}_{ij} = \mathcal{B} \cdot \log_2(1 + \text{SNR}_{ij}) \quad (14)$$

- $\mathcal{B} \rightarrow$ bandwidth allocated for the communication link between the UAVs
- $\text{SNR}_{ij} \rightarrow$ signal-to-noise ratio of the channel

Reputation

- Debit-credit reputation computation
- Offloading a task or the unsuccessful execution of an offloaded task is considered a debit from task UAV
- Successfully executing an offloaded task is considered a credit to the service UAV
- *The Reputation score (R) of a UAV is defined as a weighted sum of the current reputation score and debit-credit score available*

Reputation

- R_j of U_j^S after completing task t :

$$\mathcal{R}_j^t = (\mathcal{A}_{ij}^t \cdot \mathcal{R}_i^{t-1} - \eta_t) + \mathcal{R}_j^{t-1} \quad (7)$$

- R_i of U_i^T :

$$\mathcal{R}_i^t = \mathcal{R}_i^{t-1}(1 - \mathcal{A}_{ij}^t) \quad (8)$$

- $\eta_t \rightarrow$ boolean value that indicates the truth of unsuccessful execution

$$\eta_t = \begin{cases} 1, & \text{if } t \text{ is failure} \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

- $\mathcal{A}_{ij}^t \rightarrow$ award: it is the fraction of reputation score transferred from task UAV to service UAV upon completion of task t
- R_i and $R_j \rightarrow$ current reputation scores

Reputation

- The resources available at each UAV play an important role in computing the initial reputation
- $\mathcal{R}_j^0 = \tau \times (f_j \cdot M_j)$ (10)
- $f_j \rightarrow$ clock speed
- $\tau \rightarrow$ time constant
- τ is defined as the time taken to execute a task with the number of instructions per unit of data equal to the clock speed of the CPU
- $\tau = 1\text{s}/\text{bits}$

Reputation

- Property: *The total reputation of the system, expressed as sum of the reputation scores of each UAV, is always conserved*

Proposed Methodology

1. Trust Evaluation
2. **Service UAV Selection**
3. Dynamic Score Transfer (using Rubinstein Bargaining Game)

Service UAV Selection

- Algorithm 1 describes the process for selecting service UAVs for task offloading
- Input:
 - Set of all service UAVs, U^S
 - Set of all task assignments, T
- Output:
 - Set of assigned service UAVs, U_{ass}^S
- For each task, the algorithm evaluates each service UAV's eligibility and computes trust between task and service UAV
- The UAV with the highest trust score not already assigned is selected

Algorithm 1

INPUTS:

- 1: $\mathbb{T} = \{\mathbb{T}_1, \mathbb{T}_2, \mathbb{T}_3, \dots, \mathbb{T}_k\}$ $\triangleright$ set of all task assignments
- 2: U^S $\triangleright$ Set of all service UAVs

OUTPUT: U_{ass}^S $\triangleright$ Set of assigned service UAVs

PROCEDURE:

- 1: Initialize an empty set of U_{ass}^S
- 2: **for** $i \leftarrow 0$ to k **do**
- 3: $\zeta = 0$
- 4: **for** $j \leftarrow 0$ to $n(U^S)$ **do**
- 5: Compute $e(U_j^S, t_i, U_i^T)$ using Eq. (2)
- 6: **if** $e(U_j^S, t_i, U_i^T) = 1$ **then**
- 7: Compute $\mathcal{R}_j$ using Eq. (8) or (10) (If it is the first trial)
- 8: Compute $\mathcal{D}_{ij}$ using Eq. (14)
- 9: Compute ζ_{ij} using Eq. (1)
- 10: **if** $\zeta_{ij} > \zeta$ and $U_j^S \notin U_{ass}^S[i]$ **then**
- 11: $\zeta \leftarrow \zeta_{ij}$
- 12: $U_{ass}^S[i] \leftarrow U_j^S$
- 13: **end if**
- 14: **end if**
- 15: **end for**
- 16: **end for**
- 17: return U_{ass}^S

Fallback Mechanism

- When a task is offloaded to the most trusted UAV, but the transmission fails due to communication errors, network congestion, or UAV mobility, TOFU dynamically selects the next most trusted UAV for execution
- This fallback mechanism ensures that task execution continues within the allowed latency constraints
- If the retransmission and execution exceed the maximum permissible delay, the task is considered failed, and the reputation of the failing UAV is adjusted accordingly

Proposed Methodology

1. Trust Evaluation
2. Service UAV Selection
3. **Dynamic Score Transfer (using Rubinstein Bargaining Game)**

Dynamic Score Transfer

- Two players: task UAV and service UAV
- The reputation score is a transferable quantity
- If a UAV U_i^T offloading a task t to U_j^S , it must transfer a fraction of its score to the other UAV as the award A_{ij}^t for executing the task offloaded

Rubinstein Bargaining Model

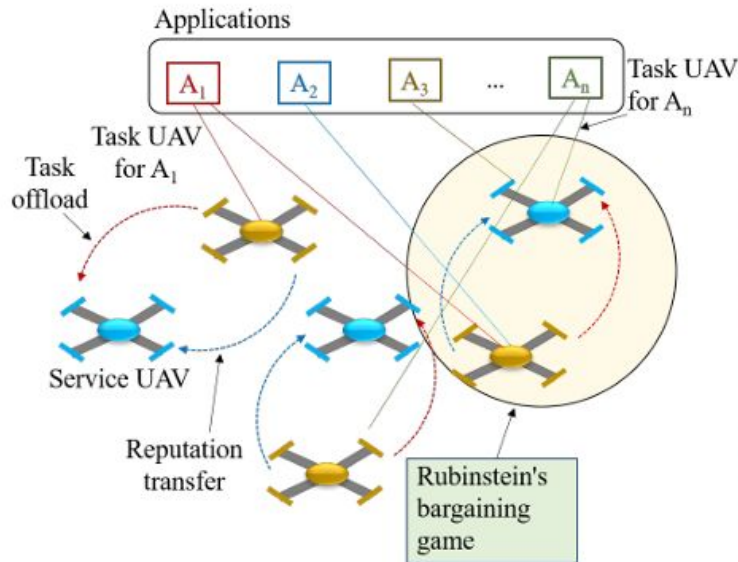
- Game theoretic approach
- Justification for using this model:
 - Helps to determine the reputation score transfer from task to service UAV
 - It also ensures a fair and strategic trust distribution
 - It dynamically adjusts the reputation distribution based on actual performance, thereby enhancing the robustness and trustworthiness of the UAV network

Rubinstein Bargaining Model

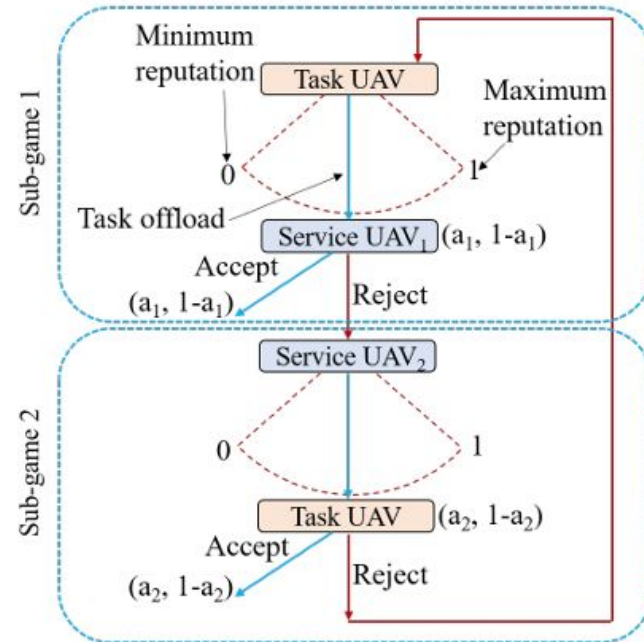
- The task UAV transfers a fraction of its reputation as an award to a service UAV for completing the task
- The task and service UAVs participate in Rubinstein's bargaining game
- Task UAV offers its fraction of reputation $(1 - a_1)$ to service UAV and keeps a_1 as its own reputation
- The service UAV can either:
 - Accept this offer or
 - Respond with a counteroffer, proposing to keep a larger share of the reputation $(1 - a_2)$ while offering a_2 to the task UAV, such that $(1 - a_2) > (1 - a_1)$ and $a_2 < a_1$
- If the task UAV is satisfied with a_2 , it offloads the task to the service UAV
- Alternatively, the task UAV tries to bargain again with the same or another service UAV

Rubinstein Bargaining Model

System Architecture



Rubinstein Bargaining Game in TOFU



Discounting Factors

- Discount factor δ_i for task UAV U_i^T
- Discount factor δ_i for service UAV U_i^S
- Based on respective reputation scores
- The discount factor is inversely related to the UAV's reputation
- UAVs with lower reputation scores can be more patient and gain higher rewards for executing offloaded tasks
- Promotes participation from lower-reputation UAVs

$$\delta_i^T = \frac{\sum_k \mathcal{R}_k^0}{\mathcal{R}_i^T} \quad (18a)$$

$$\delta_j^S = \frac{\sum_k \mathcal{R}_k^0}{\mathcal{R}_j^S} \quad (18b)$$

Subgame Perfect Nash Equilibrium (SPNE)

- The award offered by U_i^T to U_j^S is computed through Subgame Perfect Nash Equilibrium (SPNE) of the bargaining game played by them
- Theorem: *In the formulated UAV offloading game, a Subgame Perfect Nash Equilibrium (SPNE) exists that ensures both the task and service UAV reach an optimal agreement on the fraction of the reputation score to be transferred*

- $$V = \frac{(1 - \delta_j^S)}{(1 - \delta_i^T \delta_j^S)} \quad (27)$$

Subgame Perfect Nash Equilibrium (SPNE)

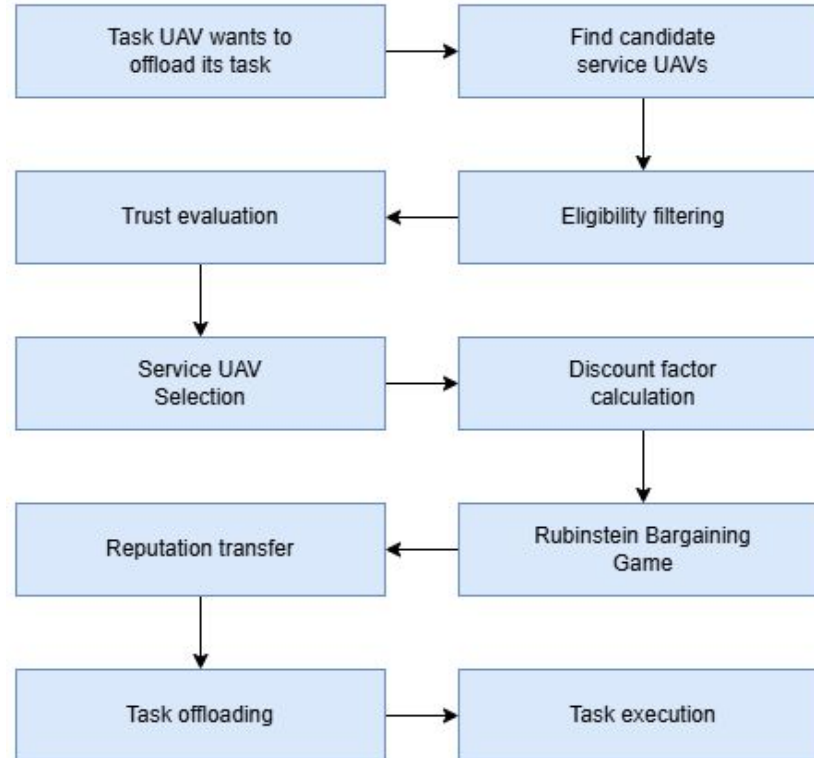
- U_i^T should offer $(V, 1 - V)$ to U_j^S to attain equilibrium
- Once the equilibrium offer (SPNE) is reached, bargaining stops immediately
- Rejecting this offer makes both UAVs worse off

Subgame Perfect Nash Equilibrium (SPNE)

- The equilibrium payoff values show that the final outcome depends on the discounting factors
- A higher discount factor means the player values future payoffs more, leading to a slower but still guaranteed convergence
- In practice, after a few rounds of negotiation, the offers reach a steady state, confirming that the game settles into a predictable and fair division of utility

- $$\mathcal{A}_{ij}^t = \frac{(1 - \delta_j^S)}{(1 - \delta_i^T \delta_j^S)} \quad (28)$$

TOFU as a flowchart



Performance Analysis

- **Reduced Delay and Energy:** TOFU reduces task delay by up to 1.74% and energy consumption per task by 2% compared to existing methods
- **Lower Failure Rates:** The scheme decreases the task failure rate by approximately 9% through its dynamic trust evaluation and fallback mechanisms
- **Energy and Reliability:** Simulations show that UAVs with higher residual energy consistently serve more tasks, whereas clock speed and memory only significantly influence the initial "bootstrapping" phase of reputation
- **Reputation Conservation:** The mean reputation of the entire system remains constant across all trials, confirming that reputation points are strictly transferred between pairs rather than created or lost

Performance Analysis

- **Scalability and Participation:** TOFU achieves the highest fraction of participating UAVs compared to DeTTO, DPTO, and SOA because its bargaining model incentivises lower-reputation drones to join the network
- **Environmental Impact:** Energy consumption increases with higher data volumes and a larger number of tasks, but it is not significantly affected by the total number of UAVs in the system
- **Service Area Constraints:** As the service area expands, the fraction of eligible UAVs decreases because drones become more dispersed, leading to lower data transmission rates

Complexity Analysis

- The overall algorithm complexity is $O(k \cdot n(U^S))$, where k is the number of tasks and $n(U^S)$ is the number of available service UAVs